Final Assignment Report - Machine Learning Engineering 2026

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1. CNNs and ViTs

In the first stage of our setup, we froze the entire **ResNet-18** convolutional base to use its pre-learned filters (pre-trained on *ImageNet*), which are highly effective at capturing generic visual features, implementing a "fast feature extraction" and trained a new classification head on our medical dataset exclusively.

The model initially achieved a peak validation accuracy (of approximately 78%) between epochs 8 and 13. However, the validation loss was volatile during the training, it remained extremely "bumpy" and failed to stabilize. This behavior pointed out that our initial learning rate of 10^{-2} was too aggressive, causing the optimizer to overshoot the local minimum and oscillate, rather than converge.

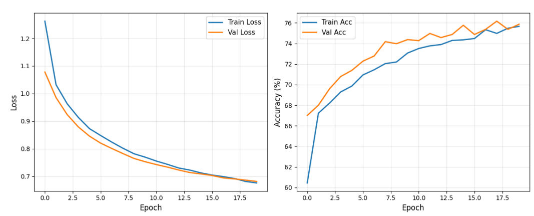

Fig. 1. Loss and accuracy curves after reducing the learning rate to 10^{-4} (ResNet-18).

To fix this, we revised the optimization strategy by lowering the learning rate to 10^{-4}. As evidenced by fig. 1, this adjustment resulted in a significantly smoother and more stable convergence trajectory. The training and validation curves aligned more closely, indicating that the model was effectively learning without the previous volatility.

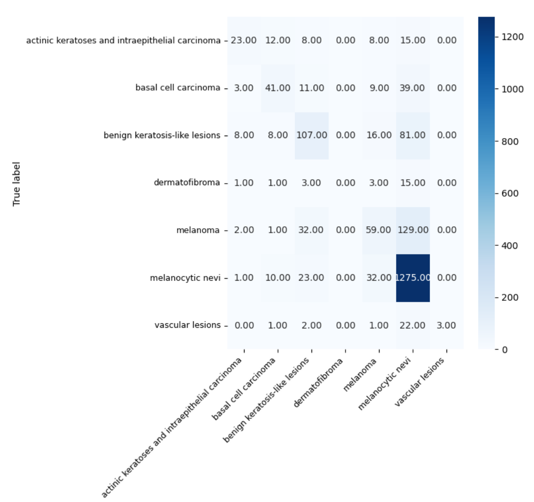

Fig. 2. Confusion matrix of the baseline ResNet-18 model.

Despite the improved stability, evaluating this baseline model confirmed the expected challenges of working with a heavily asymmetric dataset. A more in-depth analysis of the confusion matrix in fig. 2 displayed that the model achieved high global accuracy by simply defaulting to the "melanocytic nevi" majority class. This led to insufficient results for minority classes: 65% of "dermatofibroma", 76% of "vascular lesions", and 58% of "melanoma" cases were misclassified as "melanocytic nevi". In a clinical environment like ours, a false negative for melanoma is the highest-cost error, as it delays essential treatment.

To address this imbalance and prioritize diagnostic safety, we pivoted to tracking melanoma recall as our primary metric to directly minimize critical FN. Then, for the same reason, we tested two methods: a Weighted Random Sampler (WRS) and a weighted Cross-Entropy loss. We ultimately focused on the Weighted Random Sampler to ensure balanced mini-batches. We also implemented a data augmentation pipeline, including random rotations, flips, and resizing.

While the WRS improved the class distribution, the global accuracy decreased. However, this trade-off was necessary to improve the true positive rate for the melanoma class. By intentionally sacrificing global accuracy, we reduced the risk of missing malignant cases.

Finally, to adapt to the specific morphological nuances of medical skin imaging, we transitioned to a "partial fine-tuning" paradigm. We unfroze the final convolutional blocks of the ResNet-18 base. To mitigate overfitting during this phase, we applied L2 regularization (`weight_decay=1e-06`) in the optimizer, and used different learning rates: a low rate for the unfrozen convolutional layer and a higher rate for the head.

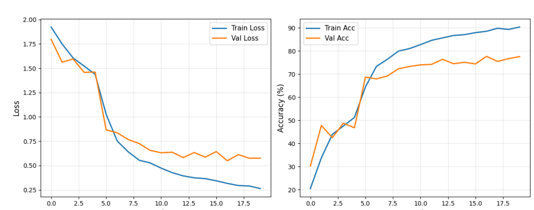

Fig. 3. Loss and accuracy curves of the partially fine-tuned ResNet-18 model.

Despite a little overfitting (fig. 3), this final configuration produced a model that better separates classes. By directly targeting the melanoma recall, we successfully dropped the FN rate from 74% in the base model to 30%, marking a substantial improvement in clinical safety (fig. 4).

For the Vision Transformer (**ViT-B/16**) architecture, we followed a training workflow focused on hyperparameter optimization and class-imbalance mitigation, mirroring the logic applied to the ResNet-18 but adjusted to the specific needs of

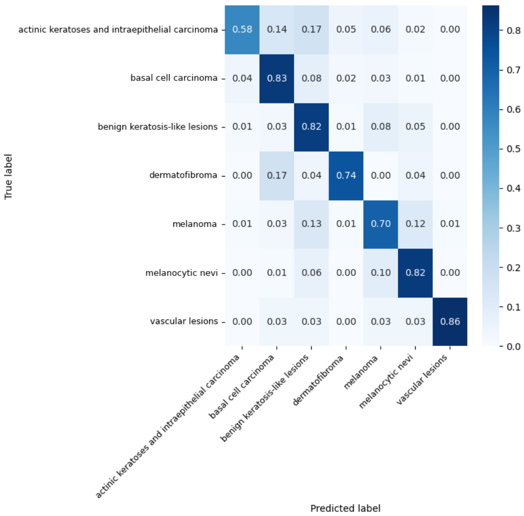

Fig. 4. Normalized confusion matrix of the final fine-tuned ResNet-18 model, showing a significant reduction in melanoma FN.

attention-based models.

LR	Tr. Acc.	Val. Acc.	Epoch	Rec. Mel.	Rec. Nev.
10^{-2}	89.18%	80.76%	14	32%	95%
10^{-3}	84.62%	81.66%	14	50%	95%
10^{-4}	75.91%	76.07%	20	57%	94%

Table 1. ViT-B/16 performance across learning rates.

Unlike the ResNet, the ViT demonstrated a strong sensitivity to the optimization scale. As documented in section 1, higher learning rates failed to provide steady generalization initially. We selected our baseline model using a lower learning rate of 10^{-4}. This rate provided stable convergence, reaching a validation accuracy of approximately 78%.

However, evaluating the confusion matrix revealed the same fundamental issue encountered previously: the model achieved high global accuracy by heavily biasing toward the “melanocytic nevi”, resulting in an unacceptably low “melanoma” recall of only 35%.

To correct this bias and prioritize diagnostic safety, we used the same data augmentation pipeline (random rotations, flips, and random cropping). We also compared two mitigation strategies: Weighted Cross-Entropy Loss (WCEL) and a Weighted Random Sampler (WRS). Because transformers lack the inductive biases of CNNs, this augmentation was critical to prevent misclassification on the minority classes. We evaluated these strategies across both 10^{-3} and 10^{-4} learning rates to observe their interaction with the optimizer’s step size.

As documented in section 1, the application of these strategies successfully shifted the model’s decision boundary away from the majority class. While the 10^{-3} configurations yielded higher recall metrics, they proved too volatile for deeper parameter optimization. Consequently, we chose the 10^{-4} configuration with WCEL and data augmentation (fig. 5) to move forward.

Finally, we transitioned to a “partial fine-tuning” strategy by unfreezing deeper blocks of the ViT architecture, applying differential learning rates, and utilizing weight decay of 10^{-6}. This final configuration yielded the most clinically viable results. By successfully pushing the true positive rate for melanoma to 78% while maintaining proportional performance across other diagnostic varieties.

Method (LR)	Train Acc.	Val Acc.	Rec. (M)	Rec. (N)
WCEL (10^{-3})	73%	72%	61%	75%
WRS (10^{-3})	79%	73%	63%	76%
WCEL (10^{-4})	68%	64%	55%	68%
WRS (10^{-4})	65%	63%	54%	69%

Table 2. ViT-B/16 performance comparing class-imbalance mitigation strategies across learning rates.

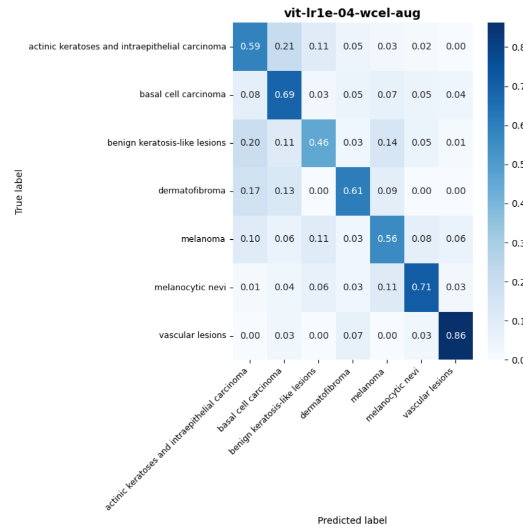

Fig. 5. Normalized confusion matrix of the ViT-B/16 model using WCEL and data augmentation at 10^{-4}, showing the targeted shift in the decision boundary.

When comparing ResNet-18 and ViT-B/16 with this imbalanced dataset, global accuracy is misleading; the true performance metric is melanoma recall. Both baseline models initially reached $\approx$ 78% accuracy by predicting the “melanocytic nevi” majority class. However, after applying class-imbalance mitigations (WRS for ResNet; WCEL for ViT) and partial fine-tuning, ViT-B/16 achieved superior diagnostic safety. The ViT reached a 78% melanoma recall, outperforming ResNet-18’s 70%, demonstrating a better capacity to separate morphologically similar classes. However, the transformer needed a longer training period, with ResNet-18 being vastly more computationally efficient ($\approx$ 236 sec. for CNN; $\approx$ 449 sec. for ViT), making it a better choice for deployment.

2. Model interpretability

Studying how the two different models perceived the images using Grad-CAM (for ResNet-18) and attention rollout (for ViT-B/16), we apprehend that the ResNet-based model demonstrates robust feature extraction by producing a uniform activation map across the entire lesion. On the other hand, the ViT clearly indicates the limitations of the transformer architecture in this context. It focuses predominantly on high-frequency contrast variations rather than synthesizing the lesion’s global morphology (fig. 6).

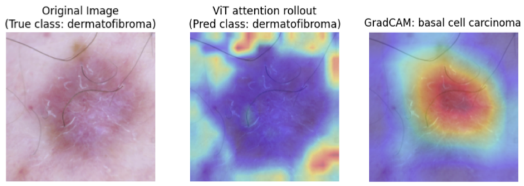

Fig. 6. An image sample where ViT correctly identifies the class but the CNN does not.

For example, even in low-confidence predictions, the CNN successfully maintains a contiguous spatial focus, mapping the full extent of the structural anomaly despite the model's uncertainty. Furthermore, regarding robustness to artifacts like body hair or surgical markings, the CNN's local pooling operations and convolutional filters render it robust to structural noise, allowing it to maintain emphasis on the underlying pathology.

Contrarily, while the ViT leverages its global receptive field to integrate broader contextual information, it often fails to isolate the clinical region of interest. Because it isn't built to naturally group neighboring pixels together (like the CNN is), its focus gets easily dispersed. Instead of looking at the solid shape of the lesion, it gets distracted by the background or tricked by random details, like focusing directly on strands of hair rather than the skin problem, or concentrating on the pieces of skin where hair is absent (fig. 7).

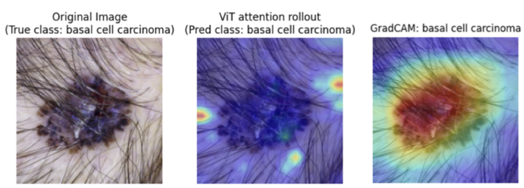

Fig. 7. Image sample containing an artifact (e.g., a hair, ink markings).

3. Fine-tuning LLMs with LoRA

In our optimization strategy, we examined the impact of four hyperparameters: rank (r), alpha (α), learning Rate (lr), and dropout. We explored a spectrum of configurations, testing ranks from $r = 8$ (minimal parameter overhead) up to $r = 64$ (higher capacity). Throughout these iterations, we set the scaling factor α both to equal to r or set to twice its value ($\alpha = 2r$). Starting from a baseline implementation scoring 35% of accuracy, the r and α values. Then, we experimented with learning rates between $5 \cdot 10^{-5}$ and $2 \cdot 10^{-3}$ and applied dropout rates from 0.05 to 0.10 to control for overfitting. Furthermore, we varied the LoRA targets, extending the adaptation from the standard attention modules (query and value) to include all linear layers, including the MLP and projection blocks.

Despite these diverse configurations and parameter coverage, the model performance consistently hit a plateau. Logs in fig. 8 showed that the training loss decreased as expected and the validation loss fluctuated in the same, narrow range. The fact that neither increasing the rank nor extending the target modules resulted in a notable accuracy increase suggests that

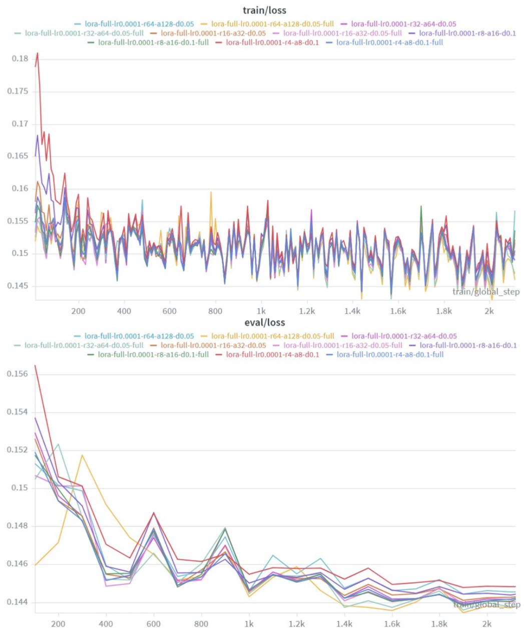

Fig. 8. Weights & Biases plots picturing the training loss and validation loss over steps (for distinct LoRA runs).

the bottleneck is not the adapter's capacity or its placement within the architecture. Instead, this indicates a fundamental limitation in the base model's reasoning or knowledge base. The minimal delta between the 35% baseline and our best 41% run suggests that the model is mainly learning formatting and simple linguistic patterns rather than achieving in-depth medical comprehension.

Ref	Content
Id 73: Most important factor in rapid progression of Bacteria resulting in pulpal necrosis is?	
GT	D
Base	The best answer is A: Host resistance.
FT	A B Perspiration is a crucial process for bacterial growth and survival
Id 36: Regeneration leads to formation of which layer of cementum: A (Cellular mixed cementum) B (Cellular intrinsic fibre cementum) C (Acellular extrinsic fiber cementum) D (Acellular afibrillar cementum)	
GT	C
Base	The best answer is A: Cellular mixed cementum.
FT	A B Persian cementum *[Non-latin characters]*.
Id 57: Common sign of occlusal trauma (TFO) is: A (Tooth mobility) B (Fractures of cusps) C (Resorption of alveolar ridge) D (Widening of P.D ligament)	
GT	A
Base	The best answer is A: Tooth mobility.
FT	A.

Table 3. Sample analysis comparing base model outputs with fine-tuned outputs.

An analysis of random samples from the test set, partially reported in table 3, reveals that the base model demonstrates a tendency toward explanatory generation. It often provides a rationale for its choice, which can be helpful for user understanding. The "fine-tuned" model is significantly more straightforward, often providing only the letter of the correct option. However, the fine-tuning process appears to have introduced forgetting and noise. For example, it generates non-existent medical terms, such as "Persian cementum" and the output is heavily contaminated with multilingual noise, including non-Latin phrases/characters.

To address the limitations of the fine-tuned model (specifically, hallucinations and multilingual noise) and to further improve the performance, Retrieval-Augmented Generation (RAG) should be incorporated. By integrating a RAG mechanism that queries verified medical databases, the system can base its outputs on external data rather than relying solely on internal parametric memory. This architectural shift would reduce the model's susceptibility to noise while potentially improving accuracy beyond the current 40% threshold.

4. RAG pipeline

Our first approach was to use larger text chunks, assuming that providing more complete information would naturally improve retrieval quality. However, early tests quickly showed that this assumption did not hold. With 512 token chunks, the model often struggled to extract specific facts and repeatedly ended up forgetting the original query focusing, instead, on summarizing the context it was given. To address this limitation, we also experimented with a larger $8k$ context window, trying to compensate for the information overload by giving the model more room to process and organize the retrieved content. In practice, this did not solve the issue. Performance dropped noticeably, and the model would sometimes stall or produce fragmented responses, indicating that it was struggling to synthesize the available information.

After several iterations, a clear pattern emerged: the model is extremely sensitive to the amount of context it receives, and more context does not automatically lead to better answers. Whenever the prompt became too dense, the model tended to drift away from the original instruction and produce weaker or incomplete responses. These observations led us to conclude that, for a small model like *Gemma 3 270M*, a lean and focused prompt is far more effective than a comprehensive one. We eventually settled on a configuration that consistently produced stable results:

Parameter	Selected value
Chunk size	128 tokens
Chunk overlap	20 tokens
Top sources	2 sources
Context window	4096 tokens
Max output tokens	4096 tokens

Comparing the "vanilla" model with the RAG-enhanced version highlighted several trade-offs.

Without RAG, Gemma produced fluent and natural-sounding responses, but they were often too generic to be useful. This caused the model to sound confident without actually providing meaningful medical information. The vanilla model occasionally generated credible but incorrect statistics, which is unacceptable in a healthcare-related application. Introducing RAG significantly reduced this issue, as the retrieved documents acted as a factual anchor and guided the model toward evidence-based responses.

Our experiments highlighted that the response style is largely influenced by the source material. While the vanilla model tends to use lists and long paragraphs, the RAG outputs were very focused and concise, often directly mirroring the input style. In a few cases, the retrieved documents had formatting issues such as split words or irregular spacing. Interestingly, the model tended to replicate these errors in its own output, generating responses with fragmented words and excessive spacing.

Occasionally, the model returned an empty response, even when the retrieval step worked correctly, and relevant documents were available. From our observations, this appeared to happen when the system prompt and the retrieved context became too large. At that point, the model simply failed to generate output. A possible fallback strategy to keep the system functional would be to report the error to the user and offer an option to forward the inquiry directly to a medical professional. This approach minimizes the workload for doctors by screening out trivial or repetitive questions while ensuring the clinical integrity of the responses. We believe that merely presenting users with links to trusted sources may be counterproductive; without professional guidance, these sources can be challenging for non-experts to interpret, potentially leading to more confusion rather than clarity.

In conclusion, we learned that a small model performs best when it is given a controlled and focused stream of information rather than a large and complex context. Prioritizing precision over volume assures that the model maintains factual integrity while avoiding the processing bottlenecks typically associated with high-density, unrefined retrieval data.

Note. Since English is not our first language, we used AI-based tools like Grammarly to check the grammar of this report.